

Perceived risks of conventional and organic produce: pesticides, pathogens, and natural toxins.



Public risk perceptions and demand for safer food are important factors shaping agricultural production practices in the United States. Despite documented food safety concerns, little attempt has been made to elicit consumers' subjective risk judgments for a range of food safety hazards or to identify factors most predictive of perceived food safety risks. In this study, over 700 conventional and organic fresh produce buyers in the Boston area were surveyed for their perceived food safety risks. Survey results showed that consumers perceived relatively high risks associated with the consumption and production of conventionally grown produce compared with other public health hazards. For example, conventional and organic food buyers estimated the median annual fatality rate due to pesticide residues on conventionally grown food to be about 50 per million and 200 per million, respectively, which is similar in magnitude to the annual mortality risk from motor vehicle accidents in the United States. Over 90% of survey respondents also perceived a reduction in pesticide residue risk associated with substituting organically grown produce for conventionally grown produce, and nearly 50% perceived a risk reduction due to natural toxins and microbial pathogens. Multiple regression analyses indicate that only a few factors are consistently predictive of higher risk perceptions, including feelings of distrust toward regulatory agencies and the safety of the food supply. A variety of factors were found to be significant predictors of specific categories of food hazards, suggesting that consumers may view food safety risks as dissimilar from one another. Based on study findings, it is recommended that future agricultural policies and risk communication efforts utilize a comparative risk approach that targets a range of food safety hazards.